Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy only when the stated facts allow it.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

With GMm=**105 J·m**, move an object from r=**3** to r=**7 m**. **Find work** by gravity using U=-GMm/r.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: mg Δh = ½mv²; v = √(**2 g** Δh).
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest **0.8 m** above its lowest point reaches speed √(**2** gh)=**4 m/s** there with g=**10** and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

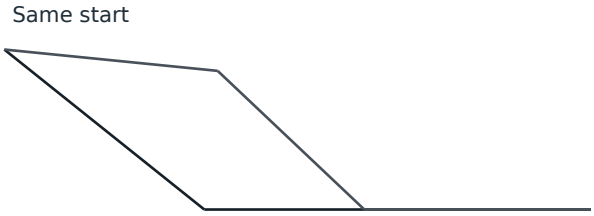
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two frictionless tracks start and end at the same heights but take different paths. **Compare final speeds** from rest.



Mark the vertical height change.

Marzano check - circle: ① ② ③

04 Apply the physics

A coaster enters height **8 m** at speed **6 m/s**.
1. **Find speed** at height **3 m** with no losses.
2. Use g=**10**.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A classmate uses total ramp length as h in mgh. **Explain how** to **name the correct measurement**.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A positive triangular pulse peaks at **60 N** and lasts **0.4 s**. It acts on a **4 kg** cart initially at **-3 m/s**. **Find impulse** and final velocity.
Show the signed impulse on the force–time graph.



Marzano check - circle: 1 2 3

07 Exit check

A cart starts from rest at a height of **5 m**. Can it reach a height of **6 m** without energy from outside? **Explain why**. **Use the energy** diagram to **show the quantities you compare**. **Label units** and unknowns.



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A frictionless coaster starts from rest at height **8 m** above the bottom of a vertical loop of radius **2 m**.
1. Use g=**10**.
2. **Find speed** and normal force per unit mass at the top; **check contact**.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3